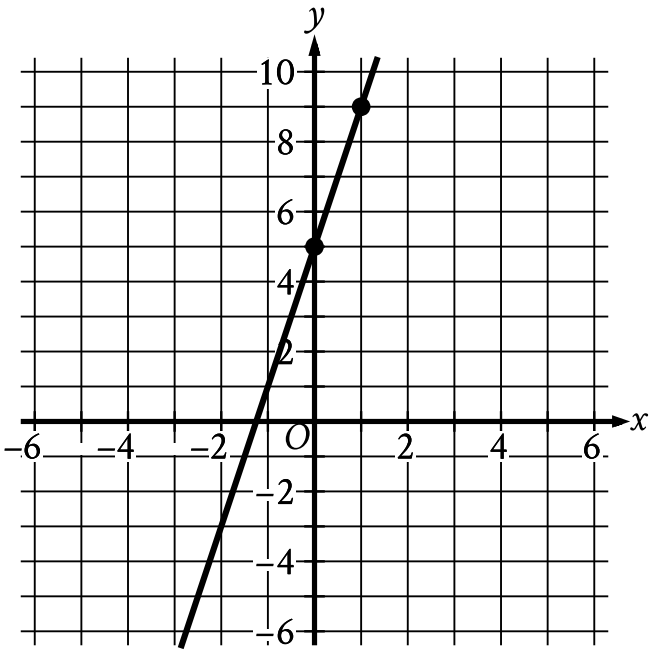


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question



Line *j* is shown in the *xy*-plane. Line *k* (not shown) is parallel to line *j*. What is the slope of line *k*?

Correct Answer: 4

Rationale

The correct answer is 4. It's given that line *k* is parallel to line *j*. It follows that the slope of line *k* is equal to the slope of line *j*. Given two points on a line in the *xy*-plane, (x_1, y_1) and (x_2, y_2) , the slope of the line can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. In the *xy*-plane shown, the points (0, 5) and (1, 9) are on line *j*. It follows that the slope of line *j* is $\frac{9 - 5}{1 - 0}$, or 4. Since the slope of line *j* is equal to the slope of line *k*, the slope of line *k* is also 4.